

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 2: Pressure Altimeters & Vertical Speed Indicators

Syllabus Source: CAE Oxford (pp. 39-78) | Pass Target: >= 90% AviationExam

1. PRESSURE ALTIMETER MECHANICS & PRESSURE DATUMS

The pressure altimeter is an absolute pressure gauge calibrated to the ICAO Standard Atmosphere (ISA: MSL pressure 1013.25 hPa, temperature +15°C, lapse rate 1.98°C / 6.5°C per 1,000 m up to 36,090 ft):

- Internal Mechanism: Contains an EVACUATED aneroid capsule (sealed with internal near-perfect vacuum). Ambient static pressure enters the instrument case. As aircraft climbs, static pressure drops; the capsule expands under internal leaf-spring tension, rotating pointer needles via jewel-pivoted magnifying gears.
- QNH (Nautical Height): Altimeter subscale set to station barometric pressure adjusted to MSL using ISA temperature lapse. Reads aerodrome elevation when on the ground; indicates ALTITUDE above mean sea level in flight.
- QFE (Field Elevation): Subscale set to actual atmospheric pressure at aerodrome reference point. Reads ZERO on ground; indicates HEIGHT above airfield runway.
- Standard Setting (1013.25 hPa / 29.92 inHg): Set when climbing through the Transition Altitude (TA). Altimeter indicates FLIGHT LEVEL (FL). Standardizes vertical separation between all cruising traffic worldwide.

2. ALTIMETER TEMPERATURE & PRESSURE ERRORS

Error Type	Atmospheric Deviation Phenomenon	Mathematical Rule & Rule of Thumb
Error Type	Atmospheric Deviation Phenomenon	Mathematical Rule & Rule of Thumb
Pressure Error	Flying from high pressure into low pressure without updating QNH subscale.	True altitude is LOWER than indicated altitude! For every 1 hPa error: altimeter misreads by approximately 27 to 30 FEET (Rule: 1 hPa = 30 ft at sea level).
Temperature Error	Flying in air colder or warmer than standard ISA (+15°C at MSL, -2°C/1,000 ft).	Rule of Thumb: 4 ft per 1,000 ft of altitude for each 1°C of ISA deviation. Cold Air: True altitude is LOWER than indicated! Hot Air: True altitude is HIGHER than indicated!

[!] EXAM TRAP (AVIATIONEXAM) - Universal Altimeter Error Mnemonic

'HIGH TO LOW OR HOT TO COLD, LOOK OUT BELOW!' - When flying from high pressure to low pressure, or from warm air to cold air: - THE ALTIMETER READS ERRONEOUSLY HIGH! (You are dangerously closer to the ground/mountain peaks than the instrument indicates)! - In sub-zero winter temperatures, published IFR approach MDA/DA and minimum sector altitudes MUST be corrected using cold temperature correction tables!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Cold Temperature Altimeter Correction

Official EASA Rule: Correction = 4 ft x (Altitude / 1,000 ft) x (ISA deviation below standard). Problem: Flying at indicated altitude 8,000 ft over mountains. OAT at 8,000 ft is -25°C. What is true altitude? Step 1: Calculate ISA temp at 8,000 ft: 15°C - (2 x 8) = -1°C. Step 2: Deviation: -25°C - (-1°C) = 24°C colder than ISA. Step 3: Correction = 4 x 8 x 24 = 768 ft. Step 4: In cold air, altimeter overreads. True Altitude = 8,000 - 768 = 7,232 ft. Rule: 'High to Low, Hot to Cold, Look Out Below!'

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Altimeter Temperature Error and True Altitude**

SCENARIO (AviationExam Classic Cold Weather Altimetry Drill): An aircraft is cruising at FL 100 with an indicated altitude of 10,000 ft on 1013.25 hPa: - The outside air temperature (OAT / SAT) is -25°C - The route passes over high mountain peaks with minimum clearance required QUESTION: Calculate the True Altitude of the aircraft, and assess the terrain clearance hazard. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Determine Standard ISA Temperature at 10,000 ft: - ISA Sea Level Temp = +15°C - ISA Lapse Rate = 2°C per 1,000 ft - ISA Temp at 10,000 ft = +15°C - (2°C x 10) = +15°C - 20°C = -5°C. Step 2: Calculate the ISA Temperature Deviation (Delta-ISA): - Actual Temp = -25°C - Delta-ISA = Actual - Standard = -25°C - (-5°C) = -20°C (Air is 20°C colder than standard ISA!). Step 3: Apply the 4 ft per 1,000 ft Rule of Thumb: - Formula: Temperature Correction = 4 ft x (Altitude / 1,000) x Delta-ISA - Correction = 4 x 10 x (-20) = -800 feet! Step 4: Calculate True Altitude: - True Altitude = Indicated Altitude + Correction - True Altitude = 10,000 ft - 800 ft = 9,200 ft! FINAL ANSWER: True Altitude is 9,200 ft. The aircraft is 800 ft lower than what the altimeter displays ('High to low, hot to cold, look out below!').

4. VERTICAL SPEED INDICATORS: VSI VS INSTANTANEOUS VSI (IVSI)

The Vertical Speed Indicator measures the rate of change of static pressure, indicating rate of climb or descent in feet per minute (fpm):

- Conventional VSI Mechanism: Contains a flexible diaphragm capsule fed directly with static pressure. The surrounding sealed case is fed through a calibrated capillary metering leak tube. In level flight, pressures equalize (0 fpm). In a climb, capsule pressure drops instantly while case pressure leaks slowly through the capillary, creating a differential pressure that deflects the needle.
- VSI Time Lag (6 to 9 Seconds): Because air takes several seconds to choke through the capillary leak, conventional VSIs suffer a 6 to 9 second lag before indicating a steady-state rate of climb/descent!
- Instantaneous VSI (IVSI): Eliminates time lag by incorporating inertia dashpot pistons (accelerometer pumps). When the aircraft enters a pitch change, inertial mass of the piston pumps instantaneous differential pressure into the capsule, giving ZERO-LAG vertical speed indication!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: An aircraft is flying at FL 100 with an indicated altitude of 10,000 ft on 1013 hPa. Outside Air Temperature (OAT) is -25°C. What is the true altitude?

- [A]: 10,000 ft
- [B]: 10,800 ft
- [C]: 9,200 ft

- [D]: 8,500 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Standard ISA temp at 10,000 ft = $15 - (2 \times 10) = -5^{\circ}\text{C}$. Actual temp = $-25^{\circ}\text{C} \rightarrow$ ISA deviation = -20°C (COLD AIR). Correction formula: $4 \text{ ft} \times (\text{Alt} / 1,000) \times \text{Delta-ISA} = 4 \times 10 \times (-20) = -800 \text{ ft}$. True Altitude = $10,000 - 800 = 9,200 \text{ ft}$!

Q2: What is the primary operational advantage of an Instantaneous Vertical Speed Indicator (IVSI) over a conventional VSI?

- [A]: It requires no static pressure connection
- [B]: It eliminates the 6 to 9 second time lag by using inertia-activated dashpot accelerometer pistons
- [C]: It indicates airspeed during cruise
- [D]: It operates on 28V DC power

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Conventional VSIs suffer a 6-9 second pneumatic metering lag. An IVSI uses acceleration dashpot pistons that pump immediate differential pressure into the capsule during maneuvers, providing instant response.

Q3: What will the pressure altimeter indicate on the runway before take-off if the subscale is set to the aerodrome elevation?

- [A]: The aerodrome elevation above MSL
- [B]: Exactly ZERO feet
- [C]: Standard flight level zero
- [D]: Pressure altitude

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. QFE is station barometric pressure at the runway reference point. Setting QFE references the altimeter to field height, reading exactly 0 ft on the runway threshold.